**BIOCHEMISTRY CUMULATIVE EXAM**

**September 13, 1997**

A liver biopsy specimen of a patient suffering from a glycogenopathy was analyzed. 0.8 Mg of glycogen isolated from the specimen was treated with phosphorylase, $\alpha$ (1$\rightarrow$ 4) to $\alpha$ (1$\rightarrow$ 4) oligo gluconotransferase and $\alpha$ (1 $\rightarrow$ 6) glucosidase in a phosphate buffer. After completion of digestion, 4.95 $\mu$ moles of glucose-1-phosphate and 0.05 $\mu$ moles of glucose were formed.

20 points each

1. Draw the structure of glycogen. You may use circles to represent a glucopyranosyl unit. Identify the reducing and nonreducing ends of the molecule and describe the branching pattern of the molecule.

2. What physiological purpose does the high degree of branching serve?

3. How does phosphorylase degrade glycogen? How do the enzymes listed above assist phosphorylase in degrading glycogen beyond the branch points?

4. What can you say about the structure of this patient's glycogen? What is the average number of glucose residues between branch points?

5. What enzyme is deficient in the patients liver? How did you reach this conclusion? What physiological consequences would you predict for this deficiency?